

Machine Learning for Data Types

Sample Exam

Part 1 – NLP (18 points)

Consider the dataset below, made of 6 documents.

	document
0	The new LMS has many features which are not ne...
0	It is not easy to navigate this LSM
0	The feature of sharing courses is useless
0	The LMS misses a feature to copy courses from ...
1	Many great and new features
1	The layout is attractive

The first 4 documents are labelled with 0, for negative and the last 2 are labelled with 1 for positive.

Assume that we have preprocessed the corpus by lowercasing the documents, removing the punctuation, and stemming all the words.

new lm mani featur need
easi navig lm
featur share cours useless
lm miss featur copi cours past year
mani great new featur
layout attract

Question 1 (3 points)

What are the dimensions (shape) of the Document-Term Matrix generated from the pre-processed dataset above?

- a. (17,6)
- b. (6,17)**
- c. (6,1)
- d. (1,6)

Question 2 (3 points)

What would be the weight of the token 'lm' in document 0 if we apply TF-IDF?

- a. 2
- b. 1
- c. 0.69 ($TF_{lm,d0}=1 * \ln(6/3)$)**
- d. 0.333

Question 3 (3 points)

Assume that we work with a logistic regression model to learn a classifier for the data above. What would be the shape of the weight vector W to be learned during training?

- a. (6,1)
- b. (17,6)
- c. (6,17)
- d. (17,1)**
- e. (1,17)
- f. (1,6)

Question 4 (3 points)

What is the total number of parameters learnable as part of this model?

- a. 6
- b. 7
- c. 17
- d. 18**

Question 5 (3 points)

Assume that we initialize all trainable parameters of the logistic regressor to 1 (weight vector and bias). What value would the model predict for the first document?

‘new lm mani featur need’.

- a. 6
- b. 1 (each word is mentioned once, 5 words in the document will yield 5, plus the bias term. We calculate the sigmoid(6), the result will be 0.997)**
- c. 0.95
- d. 0.85

Question 6 (3 points)

Assume that we learn the following weights for some of the features in our dataset.

[0.02,

3.04,

1.5,

-3.7,

-5.2,

..]

Which of these weights is more likely to be assigned to the feature ‘mani’?

- a. 0.02 (More likely to be assigned a neutral weight as the term appears in documents of both classes)**
- b. 3.04
- c. 1.5
- d. -3.7

Part 2 – Computer Vision (26 points)

Consider a dataset of 2000 flower images, with the dimensions (6,6,3).

The images are labelled as either ‘lavender’ or ‘lotus’, with an even distribution of both sets of classes.

Assume that we wish to train a FC DNN on this dataset to be able to learn a model that can classify new data points accordingly.

Question 1 (3 points)

How many input features should your FC DNN input?

- a. 2000
- b. 36
- c. **108 (6*6*3)**
- d. 216000

Question 2 (3 points)

Assume we use a FC DNN with two hidden layers, hl1 and hl2, with 16 and 8 neurons in each (respectively)

What is the shape of the weight matrices $W^{[1]}$, $W^{[2]}$ and $W^{[3]}$?

- a. **$W^{[1]}=(16,108)$ $W^{[2]}=(8,16)$ and $W^{[3]}=(1,8)$**
- b. $W^{[1]}=(16,108)$ $W^{[2]}=(16,8)$ and $W^{[3]}=(8,1)$
- c. $W^{[1]}=(108,1)$ $W^{[2]}=(16,1)$ and $W^{[3]}=(1,1)$
- d. None of the above

Question 3 (3 points)

What is the shape of the bias vectors $b^{[1]}$, $b^{[2]}$ and $b^{[3]}$?

- a. $b^{[1]}=(108,1)$ $b^{[2]}=(8,1)$ and $b^{[3]}=(1,1)$
- b. $b^{[1]}=(108,1)$ $b^{[2]}=(8,1)$ and $b^{[3]}=(1,1)$
- c. **$b^{[1]}=(16,1)$ $b^{[2]}=(8,1)$ and $b^{[3]}=(1,1)$**
- d. $b^{[1]}=(1,16)$ $b^{[2]}=(1,8)$ and $b^{[3]}=(1,1)$
- e. None of the above

Question 4 (3 points)

What is the total number of parameters to be learned when training this model?

- a. **1889**
- b. 1864
- c. 2000

d. 216000

Question 5 (3 points)

Assume we shift from a dense neural network (FC DNN) to a Convolutional Neural Network (CNN).

We first add a convolution layer, in which we use 8 kernels, of the shape (3,3). We use no padding, and the value both horizontal and vertical strides is 1.

How many feature maps will this layer output?

- a. 1
- b. 2000
- c. 8**
- d. 9

Question 6 (3 points)

What is the shape of any feature map produced by the layer above?

- a. (4,4)**
- b. (3,3)
- c. (6,6)
- d. Feature maps outputted are not of the same size

Question 7 (3 points)

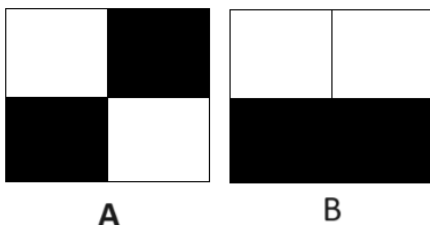
Assume that after using the one convolutional layer we defined above, we then proceed to flatten it and pass it to a DNN.

How many input feature should our DNN expect?

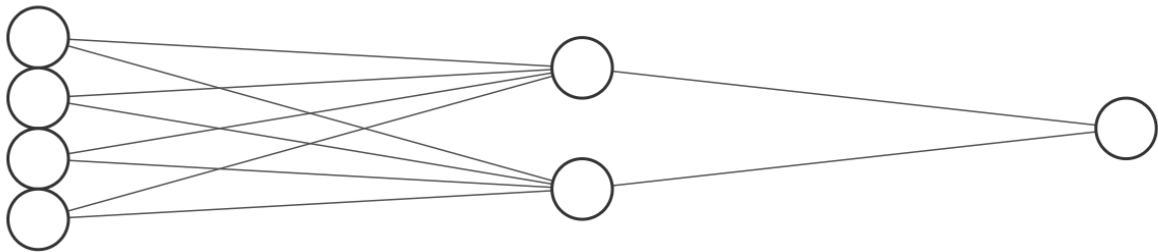
- a. 216
- b. 128 (8 feature maps, (4,4) each will be flattened)**
- c. 9
- d. 72

Question 8 (5 points)

Assume that we have a dataset of two images , a and b, with labels 1 and 0, respectively.



A black pixel is of the value 0 and a white one is of the value 1. Notice that the lower row in B has two pixels with the value 0 in each.
 Assume that we also have the following neural network architecture;



All weights and biases are initialized with the value 0.5.
 What will the network predict in the first iteration?
 For your convenience, the equations of the forward pass are available below;

$$Z^{[1]} = W^{[1]} \cdot X + b^{[1]}$$

$$A^{[1]} = \sigma(Z^{[1]})$$

$$Z^{[2]} = W^{[2]} \cdot A^{[1]} + b^{[2]}$$

$$A^{[2]} = \sigma(Z^{[2]})$$

- a. **0.78 for A and 0.78 for B**
- b. 0.78 for A and 0 for B
- c. 0 for A and 0.78 for B
- d. Both A and B will have a prediction of 0

Part 3 – TS (23 points)

Exercise 1 (5 points)

Are the following statements true or false? Explain your answer.

a. Good forecast methods should have normally distributed residuals.

- True
- **False**

Not true. It is helpful to have normally distributed residuals because it makes the calculation of prediction intervals easier, and it means that least squares estimates of parameters are also equivalent (or close to) maximum likelihood estimates. But it doesn't make the forecasts better. If the residuals are not normally distributed, one way to produce prediction intervals is to use a bootstrapped approach.

b. A model with small residuals will give good forecasts.

- True
- **False**

Not true. An over-fitted model will have small residuals (relative to other models), but will probably not give good forecasts.

c. The best measure of forecast accuracy is MAPE.

- True
- **False**

Not true. MAPE is useful in some circumstances — for example, in comparing forecasts on different scales, or with different units, and is relatively easy to understand. But it is not appropriate if the data includes zeros, or if the data has no natural zero

a. If your model doesn't forecast well, you should make it more complicated.

- True
- **False**

Not true. Some things are hard to forecast, and making a model more complicated can make the forecasts worse.

a. Always choose the model with the best forecast accuracy as measured on the test set.

- True
- **False**

Not true. Imagine the test set has only a single observation, or a very small number of observations. We don't want to select a model based on a small test set. A better approach is to use time-series cross-validation, which is based on a much larger set of test sets. Later, we will learn about the AIC statistic which is an alternative way to select a model and is often more helpful than a simple test set.

Exercise 2 (3 points)

Which of the following is not a component of time series?

- a. Seasonal variation
- b. Trend
- c. Variance**
- d. Cyclical variation

Seasonal, Cycle, and trend are three components.

Exercise 3 (3 points)

The number of cars sold by a car dealer during just 6 months in 2022 was as follows:

January	February	March	April	May	June
18	16	28	51	47	55

What is the first 3 monthly moving average?

- a. 20.67**
- b. 51.00
- c. 42.00
- d. 31.67

Just take the average of 18,16, 28.

Exercise 4 (3 points)

Data concerning events over a period of time is called a;

- a. Time Series**
- b. Random Sample
- c. Moving Average
- d. Frequency Distribution

Exercise 5 (3 points)

Find the order of the ARIMA process

$$X_t = 0.4X_{t-1} - 0.2X_{t-2} + 0.15X_{t-3} + Z_t + 0.5Z_{t-1} - 0.3Z_{t-2}$$

- a. ARIMA(2, 1, 3)
- b. ARIMA(2, 0, 3)
- c. ARIMA(3, 1, 2)
- d. ARIMA(3, 0, 2)**

Look at the coefficients of X and Z there you see t-2 and t-2 respectively which indicates that p, and q are 3 and 2 respectively.

Exercise 6 – Open Questions (3 points)

Show that a 3×5 MA is equivalent to a 7-term weighted moving average with weights of 0.067, 0.133, 0.200, 0.200, 0.200, 0.133, and 0.067.

5-term moving average:

$$z_j = \frac{1}{5}(y_{j-2} + y_{j-1} + y_j + y_{j+1} + y_{j+2}).$$

3-term moving average:

$$u_t = \frac{1}{3}(z_{t-1} + z_t + z_{t+1}).$$

Now just substituting z_j in the second equation gives you the answer.

Exercise 7 – Open Question (3 points)

Figures 1 and 2 show the result of decomposing the number of persons in the civilian labour force in Australia each month from February 1978 to August 1995.

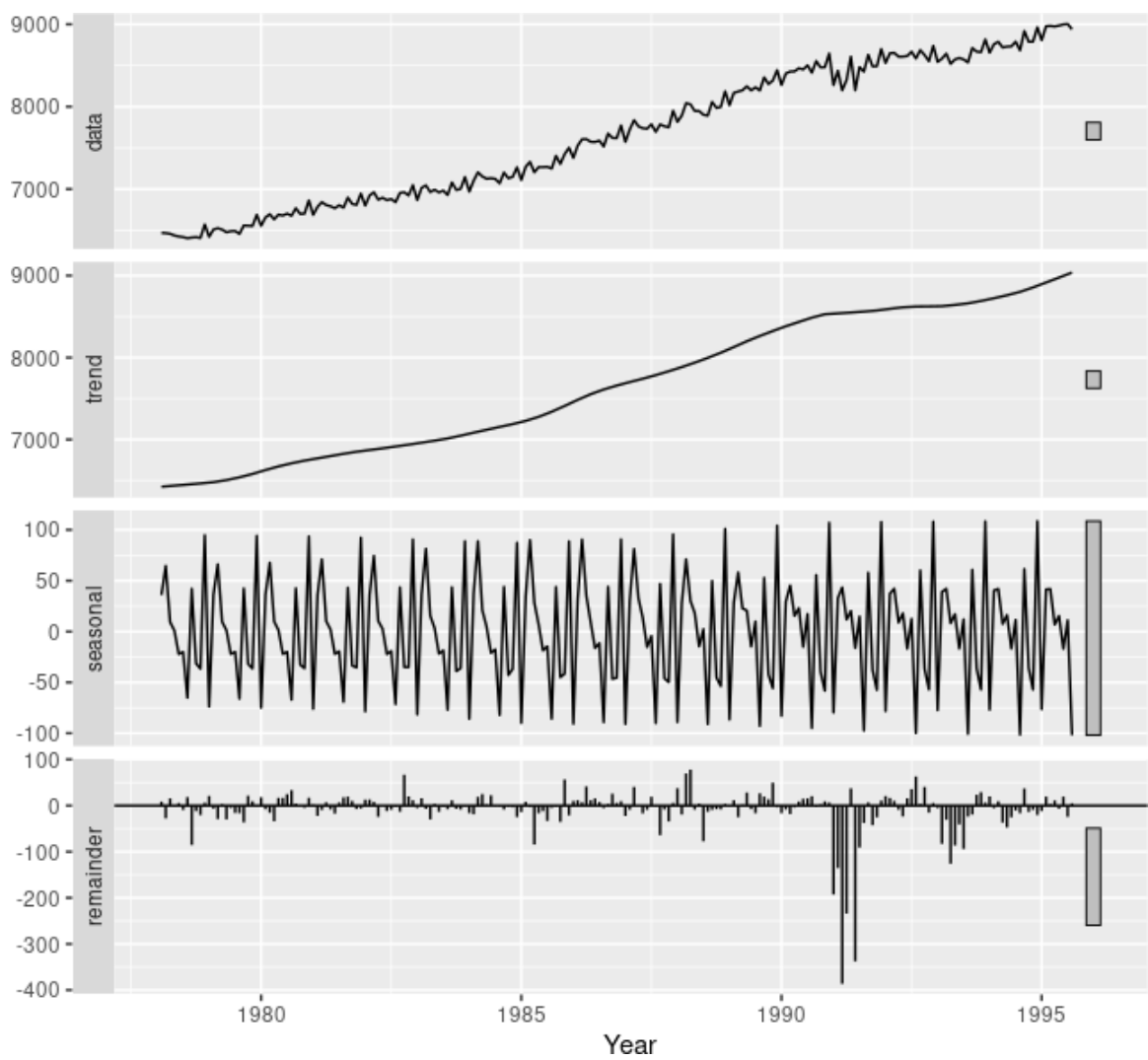


Figure 1. Decomposition of the number of persons in the civilian labour force in Australia each month from February 1978 to August 1995.

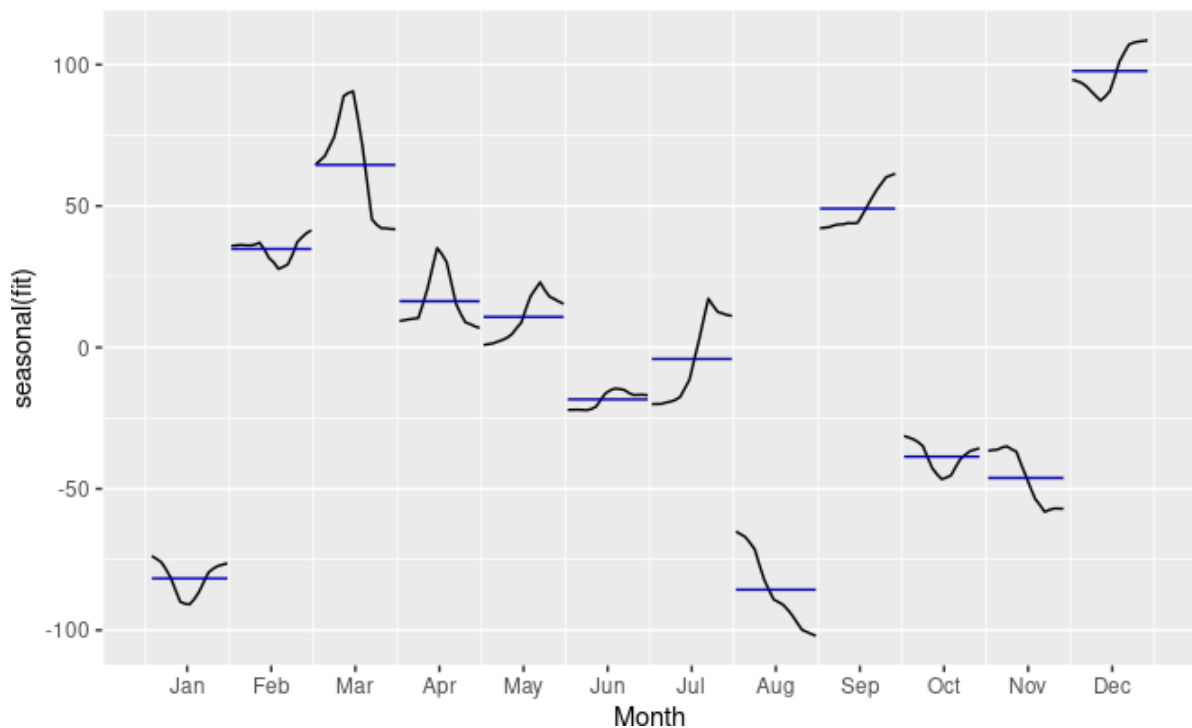


Figure 2. Seasonal component from the decomposition shown in the previous figure.

- a. Write about 3–5 sentences describing the results of the decomposition. Pay particular attention to the scales of the graphs in making your interpretation.
There is an increasing trend which is not monotonic(there is a dip between 1990-1995). There is a strong seasonality with December seeing the peak and August a dip. The strong increase in December is recent (look at Figure 2).
- b. Is the recession of 1991/1992 visible in the estimated components?
Yes notice the dip between 1990 and 1995 in the remainder component but could be observable (though not very clearly in the Trend component).